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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 06/09/2008 |
| **PI Name:** | Castaneda-Sceppa, Carmen M.D., Ph.D. | | |
| **Project Title:** | Timed Feeding and Resistance Training to Prevent Muscle Atrophy | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Nutrition, Physical Fitness, and Rehabilitation Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | UNIVERSITY | **Phone:** | 617-556-3081 |
| **Organization Name:** | Tufts University | | |
| **PI Address 1:** | Jean Mayer USDA Human Nutrition Research Center on Aging (HNRCA) | | |
| **PI Address 2:** | and Associate Professor of Nutrition | | |
| **PI Web Page:** | | | |
| **City:** | Boston | **State:** | MA |
| **Zip Code:** | 02111 | **Congressional District:** | 9 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 01-OBPR-07 |
| **Start Date:** | 09/01/2004 | **End Date:** | 12/31/2006 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Previous PI was Ronnen Roubenoff. PI changed to Castaneda-Sceppa in 2004; changed start date to reflect Castaneda-Sceppa as PI (Ed., 11/2010) | | |
| **Key Personnel Changes/Previous PI:** | Previous PI was Ronnen Roubenoff. PI changed to Castaneda-Sceppa in 2004; see Roubenoff for reports in FY2004 and earlier (this note added 2/2009). | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NCC 9-58-NPFR00301 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Muscle mass and strength, important components of health-related fitness, are significantly reduced during space flight due to a combination of factors, including the elimination of gravity and dietary deficits. Resistance training (RT) is an exercise modality that offers the hope of mitigating or reversing muscle loss induced by weightlessness. Recent studies suggest that the timing of an essential amino acid supplement around a bout of RT may augment the effect of exercise on muscle protein balance. The present study evaluates the effect of an integrated nutrition and RT approach as a countermeasure against deleterious effects of space flight on muscle health using bed rest as a model of weightlessness. The hypotheses of this study are that:<br><br>a) The combination of timed feeding (TF) of an essential amino acid (AA) supplement with RT (TFRT) is more effective than RT or TF alone as a countermeasure against bed rest induced muscle loss.<br><br>b) TFRT accelerates recovery from 28 days of bed rest compared to RT or TF alone over 14 days following strict bed rest.<br><br>c) TFRT works at least in part by modulating muscle levels of anabolic growth factors and cytokines (insulin-like growth factor-1, transforming growth factor-beta, and interleukin-15), and catabolic cytokines (interleukin -1beta, interleukin -6, myostatin, and tumor necrosis factor-alpha). |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The logistical difficulties of studying alterations in muscle during or even after space missions have stimulated the development of ground-based models, such as bed rest, to simulate weightlessness. In general, reductions in muscle strength and cross-sectional area during bed rest are very similar to those seen with space flight. For example, thirty days of bed rest produce an 18-20% decrease in isokinetic strength of the knee extensors, a 6% decrease in the knee flexors, and an 8% reduction in mid-thigh muscle CSA. Additionally, it has been shown that knee extensor strength, whole muscle cross-sectional area, and vastus lateralis fiber cross-sectional area decreases by 25-30% (various angular velocities), 14%, and 18%, respectively, after 6 wk bed rest.<br>An important question relevant to space flight conditions, as well as to many clinical situations resulting in prolonged bed rest, is the extent of loss during immobilization in addition to the length of recovery required after a prolonged stay in space (or after bed rest post-surgery). After prolonged bed rest, significant deficits in muscle strength and muscle cross-sectional area may persist for weeks, suggesting that reambulation is not enough to recover function in a timely manner. The use of concentric and eccentric actions has resulted in strength preservation or enhancement, maintenance of muscle protein synthesis, and prevention of fiber atrophy. The proposed study is the first that we know of to propose to enhance this effect using timed feedings. An understanding of the recovery process can help us design better rehabilitation strategies since recovery in the absence of rehabilitation is incomplete and muscle function is compromised even weeks post-intervention.<br><br>The earth based applicability of this project is to develop an integrated nutrition and exercise countermeasure against these deleterious effects of immobilization. However, the potential of exercise and nutritional interventions to minimize the loss of muscle function and exercise capacity, and the optimal combination of these modalities, are not known. Resistance training is one modality that offers the hope of mitigating or reversing muscle loss induced by weightlessness. Recent studies suggest that the timing of feedings around a bout of resistance training may augment the effect of exercise on muscle protein balance. This is the very question we are investigating. |
| **Task Progress:** | - Strategies for effective recruitment and retention of study subjects have resulted in enrollment of 36 subject in the study. Thirty one of these subjects have completed the study successfully and five dropped out due to time constraints or personal problems not related with the study.<br>- We continue with monthly conference calls with NSBRI study investigators with team leader, Dr. Joanne Lupton. |
| **Bibliography Type:** | Description: (Last Updated: 10/30/2019) |
| **Abstracts for Journals and Proceedings** | Castaneda C, Cloutier G, Cadena S, Pearlman J, Brooks N, Layne J, Roubenoff R. "Resistance training and timed amino acid supplementation protects against the loss of muscle mass and strength with disuse." Experimental Biology Meeting, San Francisco, Calif., April 1-5, 2006.<br>Experimental Biology, Abstract Book, April 2006. , Apr-2006 |
| **Abstracts for Journals and Proceedings** | Eksir F, Castaneda-Sceppa C, Cloutier G, Dallal G, Thomas A, Roubenoff R, Meydani SN. "Effect of bed rest with amino acid supplementation on cell-mediated immune response in healthy men: Indications for immune status of astronauts." Experimental Biology Meeting, Washington, DC, April 28-May 2, 2007.<br>Experimental Biology, Abstract Book, April 2007. , Apr-2007 |
| **Articles in Peer-reviewed Journals** | Brooks N, Cloutier GJ, Cadena SM, Layne JE, Nelsen CA, Freed AM, Roubenoff R, Castaneda-Sceppa C. "Resistance training and timed essential amino acids protect against the loss of muscle mass and strength during 28 days of bed rest and energy deficit." J Appl Physiol. 2008 Jul;105(1):241-8. Epub 2008 May 15. PMID: 18483167 , Jul-2008 |
| **Articles in Peer-reviewed Journals** | Brooks NE, Cadena SM, Vannier E, Cloutier G, Carambula S, Myburgh KH, Roubenoff R, Castaneda-Sceppa C. "Effects of resistance exercise combined with essential amino acid supplementation and energy deficit on markers of skeletal muscle atrophy and regeneration during bed rest and active recovery." Muscle Nerve. 2010 Dec;42(6):927-35. Epub 2010 Oct 6. http://dx.doi.org/10.1002/mus.21780 ; PubMed PMID: 20928906; PubMed Central PMCID: PMC2991371 , Dec-2010 |
| **Articles in Peer-reviewed Journals** | Smith SM, Castaneda-Sceppa C, O'Brien KO, Abrams SA, Gillman P, Brooks NE, Cloutier GJ, Heer M, Zwart SR, Wastney ME. "Calcium kinetics during bed rest with artificial gravity and exercise countermeasures." Osteoporos Int. 2014 Sep;25(9):2237-44. Epub 2014 May 27. http://dx.doi.org/10.1007/s00198-014-2754-x ; PubMed PMID: 24861908; PubMed Central PMCID: PMC4521405 , Sep-2014 |

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| **Articles in Peer-reviewed Journals** | Brooks NE, Cadena SM, Cloutier G, Vega-López S, Roubenoff R, Castaneda-Sceppa C. "Influence of exercise on the metabolic profile caused by 28 days of bed rest with energy deficit and amino acid supplementation in healthy men." Int J Med Sci. 2014 Sep 21;11(12):1248-57. https://doi.org/10.7150/ijms.9694 ; PubMed PMID: 25317071; PubMed Central PMCID: PMC4196126 , Sep-2014 |